

Exercise Solution

2.1 Write the truth table of the following logic functions.

$$(b) \quad F = \overline{A}\overline{B} + \overline{B}C + C\overline{D} + C\overline{A} + ABCD$$

$$(c) \quad F = (\overline{A} + B + \overline{C}) \cdot (\overline{A} + \overline{B} + D) \cdot (B + C + \overline{D}) \cdot (A + B + C + D) \cdot (\overline{A} + \overline{B} + \overline{C} + \overline{D})$$

$$(b) \quad F = \overline{A}\overline{B} + \overline{B}C + C\overline{D} + C\overline{A} + ABCD$$

A	B	C	D	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	1
1	1	1	1	1

$$(c) \quad F = (\overline{A} + B + \overline{C}) \cdot (\overline{A} + \overline{B} + D) \cdot (B + C + \overline{D}) \cdot (A + B + C + D) \cdot (\overline{A} + \overline{B} + \overline{C} + \overline{D})$$

A	B	C	D	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0

1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	0

2.2 Write the canonical “AND-OR” functions (i.e., sum of minterms) and the canonical “OR-AND” (i.e., product of maxterms) functions of the following logic functions.

$$(b) F = (V + \overline{\overline{WX}}) \cdot (V + W + X)$$

$$\begin{aligned} (b) \quad F &= (V + \overline{\overline{WX}}) \cdot (V + W + X) \\ &= (V + \overline{W} + X) \cdot (V + W + X) \\ &= M_0 \cdot M_2 \end{aligned}$$

$$\begin{aligned} F &= (V + \overline{\overline{WX}}) \cdot (V + W + X) \\ &= \overline{V} \cdot \overline{W}X + \overline{V}WX + V\overline{W} \cdot \overline{X} + V\overline{W}X + VW\overline{X} + VWX \\ &= m_1 + m_3 + m_4 + m_5 + m_6 + m_7 \end{aligned}$$

2.3 Write the simplest “AND-OR” functions of the following logic functions by K-map.

$$(b) F(W, X, Y) = \prod M(1, 5, 6, 7)$$

(b)

F W \ XY					
		00	01	11	10
0		1	0	1	1
1		1	0	0	0

$$F(W, X, Y) = \prod M(1, 5, 6, 7) = \overline{X} \cdot \overline{Y} + \overline{W}X$$

2.4 Write the simplest “AND-OR” functions of the following logic functions by K-map.

$$(a) F(W, X, Y, Z) = \sum m(1, 4, 5, 6, 7, 8, 9, 14, 15)$$

$$(c) F(A, B, C, D) = \prod M(4, 5, 12, 13, 15)$$

(a) Solution One

F		YZ			
		00	01	11	10
WX	00	0	1	0	0
	01	1	1	1	1
	11	0	0	1	1
	10	1	1	0	0

$$F(W, X, Y, Z) = \sum m(1, 4, 5, 6, 7, 8, 9, 14, 15) = \overline{W}X + XY + W\overline{X} \cdot \overline{Y} + \overline{W} \cdot \overline{Y}Z$$

(a) Solution Two

F YZ WX		YZ			
		00	01	11	10
WX	00	0	1	0	0
	01	1	1	1	1
	11	0	0	1	1
	10	1	1	0	0

$$F(W, X, Y, Z) = \sum m(1, 4, 5, 6, 7, 8, 9, 14, 15) = \overline{W}X + XY + W\overline{X} \cdot \overline{Y} + \overline{X} \cdot \overline{Y}Z$$

(c)

F		CD			
		00	01	11	10
AB	00	1	1	1	1
	01	0	0	1	1
	11	0	0	0	1
	10	1	1	1	1

$$F(A, B, C, D) = \prod M(4, 5, 12, 13, 15) = \overline{B} + \overline{C}\overline{D} + \overline{A}C$$